

Two Step Recognition of Raags in Hindustani Classical Music using Supervised Deep Learning

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Abstract—In the Indian Classical system of music, we have various raags which differ from one another based on the notes being used in them, their frequency bandwidth, the subtle intricacies of the progression of notes, etc. In [1], we have worked upon the classification of a sample of music into a thaata which is an upper-level classification simply based on the frequencies present in it. In this paper, we take the work one step ahead to recognize the raag after successful classification of the thaata. In this way, it forms a two-step process where we first identify the thaata under which the music falls, followed by which we recognize the raag which corresponds to the music in consideration.

Keywords—Raag Classification, Hindustani Classical Music, MEL Spectrogram, Supervised Learning, Deep Learning

I. INTRODUCTION

A thaata is an encapsulation of similar kinds of raags based on the notations i.e., frequencies present in it. The thaata classification system has already been proposed in [1], followed by which our next task is to detect the raag in which the music in consideration has been performed. Raag classification consists of various subtle aspects where there is a need to analyze the whole system at a more granular level. Feature extraction is an important and integral part of these studies where the accuracy of the model depends upon the relevant features that we can derive from our music and their significance. When it comes to audio signals, their features are broadly categorized into four namely:

1. Time domain features
2. Frequency domain features
3. Perceptual features
4. Windowing features

When it came to the classification of thaatas we considered various STFT, CQT, CENS, Spectral features, MFCC, etc. These features helped us in an overall upper-level classification of the samples into thaatas. For further granular-level classification, we need to extract a few other additional features from the sound samples and train our model using them.

In [1], the whole thaata classification system has been explained along with various feature engineering techniques, which is a prequel to this work. To capture the raga's melodic outline, authors in [2] have proposed a feature called Time Delay Melody Surface based on the delay coordinates. They have also presented a system in [3] to recognize raags based on their melodic phrases using pattern extraction, clustering & processing using a vector space model. The authors also provided us with the Raga Dataset from where we used the Hindustani Music Dataset for acquiring our data for this study. In [4], a spectral contrast feature has been proposed based on octaves for genre classification in music. Authors in [5] have discussed various techniques, algorithms & their complexities to enhance the spectral contrast.

A technique for the detection of changes in the harmonic contents of audio signals has been proposed by authors in [6]. The python module designed for working with Audio signals - Librosa 0.4.0 has been described by the authors in [7]. Various techniques have been reviewed, compared, and categorized by authors in [8] to detect the onset and present possible enhancements. In [9] a music generation system using Stateful Char RNNs has been proposed which will be later tweaked and integrated with this work in the next stage for real-time application purposes. The signal analysis in Hindustani Classical Music has been explained by the authors in [10]. The computational aspects of musicology in Hindustani Classical Music are discussed in [11].

In this paper, we have used various other features as discussed in [10] & [11] which address the further atomic aspects of music & used complex deep learning models to design our system.

II. RAAG CLASSIFICATION

A. The 12 Notes

Table 1 shows the notations of the 12 notes and their names which form the basis of a thaata or a raag. These 12 notes repeat themselves in the exact cyclical manner when we move up or down any number of octaves. On a frequency domain, if f is the frequency for a note in the middle octave, in the upper octave it becomes $2f$ and in the lower octave, it becomes $f/2$.

Note Notation	Full Name
Sa	Shadaj
Re ₁	Komal Rishabh
Re	Rishabh
Ga ₁	Komal Gandhaar
Ga	Gandhaar
Ma	Madhyama
Ma ⁺	Teevra Madhyama
Pa	Pancham
Dha ₁	Komal Dhaivat
Dha	Dhaivat
Ni ₁	Komal Nishaad
Ni	Nishaad

Table-1: Notes, notations, and their names

Of the notes mentioned above, there are 7 pure notes (Sa, Re, Ga, Ma, Pa, Dha, Ni) and 5 deviated notes (Re₁, Ga₁, Ma⁺, Dha₁, Ni₁). Sa and Pa are the two notes which do not deviate.

B. Shrutis

The pure notes that we see above are not uniformly separated from one another in uniform distribution. Great performers and musicologists have identified various other frequencies between any two notes in the frequency domain known as Shrutis. There is a total of 22 shrutis in an octave. Table 2 shows the note range and the number of shrutis between them.

Notes	Number of Shrutis
Sa to Re	4
Re to Ga	3
Ga to Ma	2
Ma to Pa	4
Pa to Dha	4
Dha to Ni	3
Ni to Sa ⁺	2

Table-2: Notes and the number of shrutis between them

From the above table, we can infer that Sa and Re are placed at a distance of 4 shrutis, Re and Ga at a distance of 3 shrutis, and so on. These shrutis are important as they form an integral part of a raag especially when core classical genres like Dhrupad-Dhamar are being performed.

III. SYSTEM OF RAAGS

As explained above, a thaata is an encapsulation of raags of similar notes. Once the thaata has been correctly identified using the notes and their corresponding frequencies, the next step is to come back to the time domain and analyze the patterns, elements, and checkpoints through which the notes are being used. These subtle aspects manifest themselves in various forms such as Vadi, Samvadi, Anuvadi, and Vivadi swars. For a raag, there are specific Varjit, Ansh, Grah & Nyas Swars. By the term – ‘swar’, we are referring to a note. These elements forming a flexible rule in a raag have been explained below.

Aaroh:

It is the pattern/sequence of notes through which we move up in an octave, in a given raag.

Avroh:

It is the pattern/sequence of notes through which we move down in an octave, in a given raag.

Vadi Swar:

It is the most significant and important note that is used in a raag. For example, in Raag Bhupali the Vadi Swar is Ga.

Samvadi Swar:

It is the second most significant and important note that is used in a raag. For example, in Raag Bhupali the Samvadi Swar is Dha.

Anuvadi Swar:

These are the other notes being used apart from the Vadi and Samvadi notes in a given raag. For example, in Raag Bhupali the Anuvadi swars are Sa, Re, and Pa.

Varjit Swar:

These are the notes, the used of which are forbidden in a raag. For example, in Raag Bhupali Ma & Ni are forbidden to use.

Vivadi Swar:

These are the notes which deteriorate the form of the raag but are sometimes used to beautify it. For example, in Raag Bihag, Ma⁺ used in the avroh is the Vivadi swar.

Chalan:

A chalan is the actual sequence that inculcates all the rules and limitations as discussed above and forms the whole structure of the raag. All major improvisations and compositions must be based on the chalan.

Pakad:

A pakad is just a short sequence that is also used as an identifier of the raag.

Based on these subtle aspects described above, a given piece of music may fall under the same thaata (based on notes) but may be in different raags based on the nature of their progression.

IV. THAAT-RAAG MAPPING

Based on the 10 thaats [1] – Bilawal, Kalyan, Khamaj, Kaafi, Asawari, Bhairav, Bhairavi, Marwa, Purvi & Todi, we extracted 5 individual raags from each of these thaats using the same Raga Dataset and few other sources. 100 audio samples (of 30 seconds each) of 5 raags each belonging to each thaata including the ones we used earlier for thaata classification we collected and sampled using Audacity 3.1.3. Hence, our total duration = $30 \times 100 \times 5 \times 10 = 150000$ seconds ≈ 42 hours of music data. Hence, this also takes the form of a multi-class classification problem within each thaata. The considered raags from various thaats, along with their mapping are given in the tables below:

Kaafi	Todi
Kaafi	Miyan ki Todi
Bageshri	Multani
Abhogi	Gujari Todi
Bhimpalasi	Madhuvanti
Malhar	Bairagi Todi

Table-3: Considered raags in Kaafi & Todi Thaats

Bhairav	Bhairavi
Bhairav	Bhairavi
Ahir Bhairav	Malkauns
Bibhas	Bilaskhani Todi
Gunkali	Komal Rishabh Asavari
Nat Bhairav	Kaunsi Kanada

Table-4: Considered raags in Bhairav & Bhairavi Thaats

Asavari	Bilawal
Asavari	Bilawal
Darbari	Durga
Adana	Bihag
Desi	Shankara
Jaunpuri	Deshkar

Table-5: Considered raags in Asavari & Bilawal Thaats

Purvi	Marwa
Purvi	Marwa
Gauri	Sohini
Shree	Jait
Basant	Bhatiyar
Lalit	Sazgiri

Table-6: Considered raags in Purvi & Marwa Thaats

Kalyan	Khamaj
Yaman	Khamaj
Bhupali	Rageshree
Hindol	Jhinjhoti
Kedar	Desh
Hameer	Tilak Kamod

Table-7: Considered raags in Kalyan & Khamaj Thaats

V. FEATURE EXTRACTION

To consider the types of notes and sequence in which we move up or down an octave along with the imposed limitations as explained in Section III, we have two approaches. The first approach is to consider each element individually across the duration of the sample. For example, the count of notes gives us an idea about the Vadi & Samvadi swars. The average time duration for a note gives us an idea about the structure of the raag. The onset of notes gives us an idea about the Aaroh & Avroh.

The second approach is to consider the overall nature and framework of the raag over a certain period. This can be captured using the chalan or pakad of the raag as it inculcates all the rules which must be followed by a raag. We can utilize spectrograms to capture the whole sequence in one go. Figures 1 & 2 show the spectrograms from one of the samples of Asavari and Bhairavi raags as mentioned in Section IV.

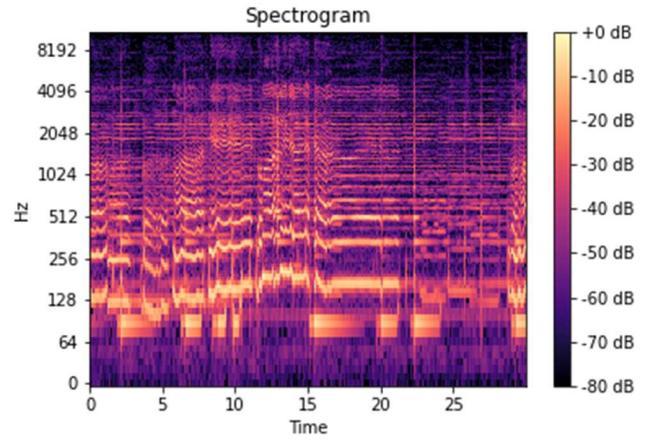


Figure-1: Spectrogram for Asavari Raag's sample

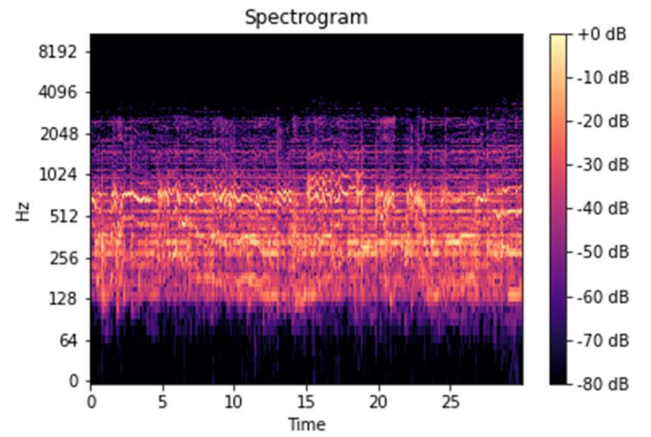


Figure-2: Spectrogram for Bhairavi Raag's sample

Mel Scale:

Humans are incapable of perceiving frequencies on a linear scale. We detect differences in lower frequencies better than that at higher frequencies. The authors in [13] proposed Mel Scale in which equal distances in pitch were perceived equally by a human. The below formula is popularly used to convert frequency (f) in hertz to mels (m).

$$m = 2595 \log \left(1 + \frac{f}{700} \right) = 1127 \ln \left(1 + \frac{f}{700} \right)$$

To consider the whole chalan of the raag in a piece of music, we generated Mel Spectrograms corresponding to each piece of music and trained our supervised model on top of that. Hence, we acquired $50 \times 5 \times 10 = 2500$ such images which have been further fed to a CNN model to acquire the subtle features from a sample of the music.

VI. MODELLING

The piece of music to be recognized is first input into the system from which the vectorial features are extracted, followed by which the features are fed into the Thaats classifier [1] to classify the given sample of music into either of the 10 thaats. After the thaats is recognized, the Mel Spectrogram for the same input file is generated which is further fed to one of the Raag Detectors ($R_1, R_2, \dots, R_{10}$) where each detector corresponds to one thaats. Finally, only one of the Raag detectors recognizes the raag to which the input piece of music corresponds. The flowchart for the whole cascaded scheme of Raag recognition has been shown in Figure 3.

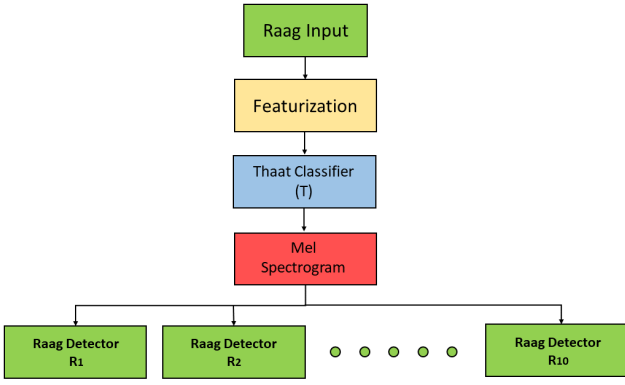


Figure-3: Two-Step Raag Detection Scheme

The spectrograms generated from the images have been processed by a convolutional neural network as shown in Figure 4, the summary for which has been shown in Figure 5.

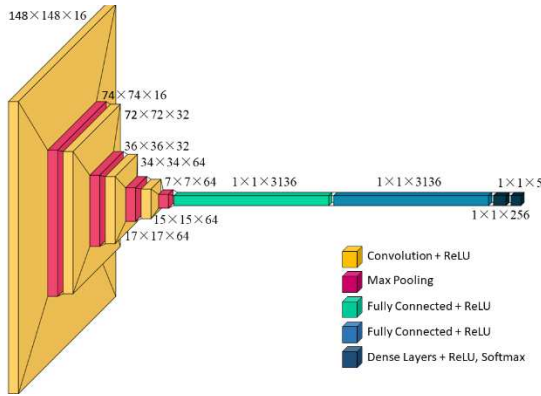


Figure-4: The CNN Model

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 148, 148, 16)	448
max_pooling2d_4 (MaxPooling 2D)	(None, 74, 74, 16)	0
conv2d_5 (Conv2D)	(None, 72, 72, 32)	4640
max_pooling2d_5 (MaxPooling 2D)	(None, 36, 36, 32)	0
conv2d_6 (Conv2D)	(None, 34, 34, 64)	18496
max_pooling2d_6 (MaxPooling 2D)	(None, 17, 17, 64)	0
conv2d_7 (Conv2D)	(None, 15, 15, 64)	36928
max_pooling2d_7 (MaxPooling 2D)	(None, 7, 7, 64)	0
flatten_1 (Flatten)	(None, 3136)	0
dropout_1 (Dropout)	(None, 3136)	0
dense_2 (Dense)	(None, 256)	803072
dense_3 (Dense)	(None, 5)	1285

=====
 Total params: 864,869
 Trainable params: 864,869
 Non-trainable params: 0

Figure-5: The model summary

The CNN model was trained individually for each of the ten detectors ($R_1, R_2, \dots, R_{10}$) with 8:2 train-test split ratios, i.e., 4000 samples for training and 1000 samples for testing the accuracy on the test set, uniformly distributed. This means that for each thaata we have 400 train samples and 100 test samples of music points.

VII. RESULTS & DISCUSSION

Individually these models were trained up to 100 epochs in batches of size 32. The loss function was kept as categorical cross-entropy whose value has been minimized, using Adam Optimiser. Each of the detectors ($R_1, R_2, \dots, R_{10}$) was tested individually. The average loss and accuracy values across all the 10 detectors over 100 epochs have been shown in figure 6. The test accuracy values across each detector have been shown in figure 7.

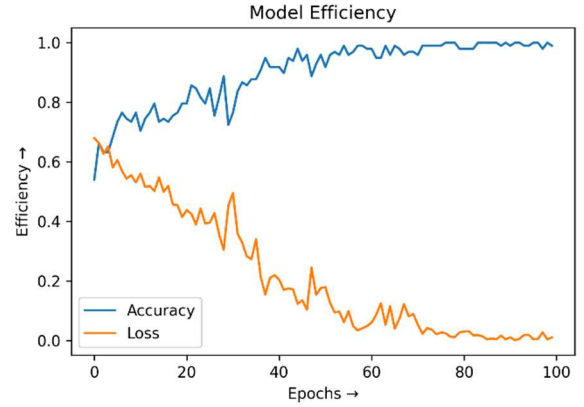


Figure-6: Final Train loss values across each detector

The lowest loss value of 0.0313 was found for Todi thaata and 0.0089 for Bhairavi thaata.

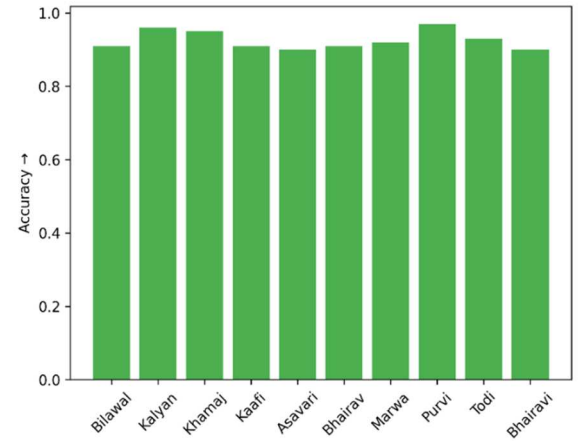


Figure-7: Test accuracy values across each detector

We acquired the highest accuracy of 97% for Purvi and Kalyan thaats and the lowest of 90% for Todi thaata on test data as shown in the figure above.

VIII. CONCLUSION

In this paper, we merged the classifier in [1] with Raag detectors corresponding to each thaata. After successful recognition of the raag, we can integrate the model with an LSTM-based Raag generation model based on the notes and sequences being used in it as mentioned in [9]. This can help remove plagiarism in the music industry up to a certain extent and help music composers come up with new ideas instead of

creating remixes and remakes from past songs in the music industry. There's a lot to discover and extract patterns from, hence why not extract subtle patterns from various sources instead of reusing the same chords and progressions repeatedly?

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